

FACULTY OF ENGINEERING, UNIVERSITY OF JAFFNA

EC4010 – DIGITAL DESIGN

LABORATORY SESSION 4

REGISTERS AND COUNTERS

REGISTRATION NUMBER:

DATE:

OBJECTIVES:

- Identify the functionalities of registers and explain how it can be implemented in digital logic design.
- Describe the functionalities of ripple and synchronous counter and design the counters using sequential logic.

APPARATUS:

- ICs 7404, 7408 and 7432
- Bread board and wires.
- LED
- Resistors

TASKS:

1. What is the difference between serial and parallel transfer? Explain how to convert serial data to parallel and parallel data to serial. What type of register is needed?
2. Design a four-bit shift register with parallel load using D flip flops. There are two control inputs: shift and load. When shift is 1, the content of the register is shifted by one bit position. New data are transferred into the register, when load is 1 and shift is 0. If both control inputs are equal to 0, the content of the register does not change.
3. Draw logic circuit diagram of a four-bit binary ripple countdown counter using flip flops that trigger on the positive edge of the clock.
4. Design a four-bit binary synchronous counter with D flip flops.